United States Patent	12402119
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Ye; Shiangrung

Multiple uplink carriers in a cell deployed in unlicensed spectrum

Abstract

Processing hardware (**134**) in a base station (**104**) can implement a method for configuring uplink communications in an unlicensed spectrum. The method includes selecting a first uplink carrier and a second uplink carrier for a cell to operate as a normal uplink carrier (NUL) and a supplementary uplink carrier (SUL), respectively, with at least one of the NUL or the SUL operating within one or more unlicensed bands the user device supports (**704**). The method further includes transmitting, to the user device, configuration data for the NUL and the SUL (**706**), including transmitting an indication that one of the carriers is the NUL, and the other carrier is the SUL, such that the user device is configured to initially transmit via the NUL. The method also includes receiving uplink data via the second uplink carrier upon the user device switching from the first uplink carrier to the second uplink carrier (**712**).

Inventors:	**Ye; Shiangrung (New Taipei, TW)**
Applicant:	**GOOGLE LLC** (Mountain View, CA)
Family ID:	**1000008779318**
Assignee:	**GOOGLE LLC (Mountain View, CA)**
Appl. No.:	**17/290751**
Filed (or PCT Filed):	**October 31, 2019**
PCT No.:	**PCT/US2019/059193**
PCT Pub. No.:	**WO2020/092787**
PCT Pub. Date:	May 07, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20210385836 A1	Dec. 09, 2021

Related U.S. Application Data

us-provisional-application US 62754415 20181101

Publication Classification

Int. Cl.: **H04W72/1268** (20230101); **H04W16/14** (20090101); **H04W72/0453** (20230101); **H04W72/23** (20230101)

U.S. Cl.:

CPC **H04W72/1268** (20130101); **H04W16/14** (20130101); **H04W72/0453** (20130101); **H04W72/23** (20230101);

Field of Classification Search

CPC: H04L (5/001); H04L (5/0098); H04W (16/14); H04W (72/0453); H04W (72/1268); H04W (72/23); H04W (76/15); H04W (76/20)

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Primary Examiner: Sinkantarakorn; Pao

Attorney, Agent or Firm: MARSHALL, GERSTEIN & BORUN LLP

Background/Summary

FIELD OF THE DISCLOSURE

(1) This disclosure relates to wireless communications and, more particularly, to configuring uplink carriers for a user device operating in a cell deployed in unlicensed spectrum.

BACKGROUND

(2) The background description provided herein is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

(3) In some cases, base stations and user devices operating in wireless communication networks can utilize both licensed and unlicensed spectrum. For example, base stations and user can utilize Licensed Assisted Access (LAA) techniques to access the unlicensed spectrum. According to the Long Term Evolution (LTE) standards, secondary cells (SCells) can operate in the unlicensed spectrum, whereas primary cells (PCell) must operate in the licensed spectrum. 5G Next Radio (NR) communication standards proposes to also deploy PCells in the unlicensed spectrum, in some cases.

(4) In addition extending unlicensed spectrum to PCells, NR supports supplementary uplink (SUL) communications in a cell to provide better coverage in the uplink direction. Unlike cells with one normal uplink (NUL) carrier and one normal downlink (NDL) carrier, such a cell includes one NUL carrier, one SUL carrier, and an NDL carrier.

(5) Further, carriers (i.e., physical channels) can occupy higher or lower portions of the spectrum. High-frequency spectrum in general corresponds to larger pathloss and larger penetration loss than low-frequency spectrum. For example, deploying a NUL in a high-frequency band reduces the coverage of the cell. On the other hand, in general larger transmissions bandwidths are available in high-frequency spectrum, which allows devices to increase data transmission rate.

SUMMARY

(6) Generally speaking, the techniques of this disclosure allow a base station to configure uplink communications from a user device via two uplink carriers, with at least one of the uplink carriers deployed in the unlicensed spectrum. One of the uplink carriers can operate as a NUL and the other uplink carrier can operate as a SUL. To take advantage of respective benefits of transmitting in low- and high-frequency spectrum, the base station can allocate one of the uplink carriers in a high-frequency transmission band and allocate the other uplink carrier in a low-frequency transmission band (e.g., with an example central frequency separation of 20 MHz or more).

(7) In operation, the user device switches from the first uplink carrier to the second uplink carrier (and, when appropriate, back to the first uplink carrier) in response to a command from a base station or upon evaluating one or both uplink carriers at the user device. More particularly, each of the user device and the base station can estimate pathloss and calculate a metric of a channel status for a carrier. A channel status metric can reflect the amount of interference, an amount of traffic, power level, etc. The user device and/or the base station can compare pathloss (which also can be a quantitative metric) and the channel status metric to threshold values to determine when the user device should switch between carriers.

(8) In some example scenarios, when the user device experiences large pathloss, or high interference, or high traffic load on the first uplink carrier, the user device switches uplink transmissions to the second uplink carrier. When the user device experiences large pathloss, or high interference, or high traffic load on the second uplink carrier, the user device switches uplink transmissions to the first uplink carrier.

(9) To mitigate race conditions and the so-called ping pong effect, where the user device and the base stations frequently arrive at conflicting decisions regarding uplink carrier selection, the user device and the base station can run timers to prevent the user device from switching again within a

certain period of time.
(10) One example embodiment of these techniques is a method in a base station for configuring uplink communications in an unlicensed spectrum, which can be executed by processing hardware. The method includes selecting a first uplink carrier and a second uplink carrier for a cell, including selecting at least one of the first uplink carrier and the second uplink carrier from within one or more unlicensed bands the user device supports. The method further includes transmitting, to the user device, configuration data for the first uplink carrier and the second uplink carrier, and receiving uplink data via the second uplink carrier upon the user device switching from the first uplink carrier to the second uplink carrier.
(11) Another example implementation of these techniques is a non-transitory computer-readable medium that stores instructions which, when executed by processing hardware, cause a base station to implement the method in the base station above.
(12) Yet another example embodiment of these techniques is a method in a user device, which can be executed by processing hardware. The method includes receiving, from a base station for a certain cell, a configuration of a first uplink carrier and a second uplink carrier, such that at least one of the first uplink carrier and the second uplink carrier are allocated in an unlicensed spectrum. The method further includes transmitting data over the first uplink carrier, detecting a condition for switching from the first carrier to the second carrier and, in response to detecting the condition, transmitting further data over the second uplink carrier.
(13) Still another example implementation of these techniques is a non-transitory computer-readable medium that stores instructions which, when executed by processing hardware, cause a base station to implement the method in the user device above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a block diagram of an example wireless communication system in which a base station can configure a user device with two uplink carriers in a same cell, and the user device can switch between two uplink carriers in a cell deployed at least partially in unlicensed spectrum;
(2) FIG. **2** is a messaging diagram of an example scenario in which the user device of FIG. **1** determines that it should switch from one uplink carrier to another uplink carrier;
(3) FIG. **3** is a messaging diagram of an example scenario in which the base station of FIG. **1** determines that the user device should switch from one uplink carrier to another uplink carrier;
(4) FIG. **4** is a messaging diagram of an example scenario in which the base station of FIG. **1** queries the user device to determine which licensed and unlicensed bands the user device supports, and configures the two uplink carriers accordingly;
(5) FIG. **5** is a flow diagram of an example method for determining whether the user device should switch from one uplink carrier to another uplink carrier, which can be implemented in the user device of FIG. **1**;
(6) FIG. **6** is a flow diagram of an example method for determining whether the user device should switch from one uplink carrier to another uplink carrier, which can be implemented in the base station of FIG. **1**;
(7) FIG. **7** is a flow diagram of an example method for configuring a user device with two uplink carriers and receiving data over these uplink carriers, which can be implemented in the base station of FIG. **1**; and
(8) FIG. **8** is a flow diagram of an example method for transmitting data over one of two uplink carriers, which can be implemented in the user device of FIG. **1**;
DETAILED DESCRIPTION OF THE DRAWINGS
(9) Some of the techniques of this disclosure can be implemented in a user device (commonly

referred to using the acronym UE, which stands for "user equipment"), and some can be implemented in a cellular infrastructure, e.g., in a base station or a CN. Generally speaking, these techniques allow a UE to dynamically switch between uplink carriers when transmitting data to a base station. As used herein, "switching" between uplink carriers can refer to moving uplink transmissions from Physical Uplink Shared Channel (PUSCH), Physical Uplink Control Channel (PUCCH), Signal Reference Signal (SRS), etc. on one carrier to the other carrier. When the base station allocates these uplink carriers in different parts of the spectrum, there may be a significant difference in various aspects of the performance of these uplink carriers, and thus the UE in different situations can select the better performing uplink carrier.
(10) These techniques are discussed below with example reference to New Radio (NR). However, in general the techniques of this disclosure also can apply to other radio access technologies (RATs).
(11) Referring first to FIG. **1**, a UE **102** can operate in an example wireless communication network **100**. As discussed below, the UE **102** can be any suitable device capable of wireless communications. The wireless communication network **100** includes an NR base station **104** connected to an EPC or 5GC **106**. The base station **104** operates as a next-generation evolved Node B (gNB) and covers an NR unlicensed (NR-U) cell **108**. In particular, the base station **104** deploys the cell **108** at least partially in unlicensed spectrum. The cell **108** can be a primary cell (PCell) or a secondary cell (SCell).
(12) The base station **104** configures the UE **102** to communicate with the base station **104** over a normal uplink carrier (NUL) **110** and a supplementary uplink carrier (SUL) **112** in the uplink direction, and over a normal downlink carrier (NDL) **114** in the downlink direction. The base station **104** associates the NUL **110** and the SUL **112** with the same cell **108**.
(13) In one example implementation, the base station **104** selects the NUL **110** from within an unlicensed band and the SUL **112** from within an licensed band. In another example implementation, the base station **104** selects the NUL **110** from within a licensed band and the SUL **112** from within an unlicensed band. In yet another example implementation, the base station **104** selects both the NUL **110** and the SUL **112** from within an unlicensed band.
(14) In some implementations, the base station **104** allocates one of the carriers **110**, **112** in high-frequency spectrum and the other one of the carriers **110**, **112** in low-frequency spectrum. For example, the base station **104** can allocate the NUL **110** in high-frequency spectrum and the SUL **112** in low-frequency spectrum. When choosing the carriers **110** and **112**, the base station **104** in various implementations uses contiguous portions of the spectrum or non-contiguous portions of the spectrum. Thus, high-frequency spectrum within which the base station **104** selects the carrier **110** can correspond to one contiguous transmission band or multiple non-continuous transmission bands, and the low-frequency spectrum within which the base station **104** selects the carrier **112** similarly can correspond to one contiguous transmission band or multiple non-continuous transmission bands.
(15) In one example implementation, the central frequency of the transmission band in high-frequency spectrum, within which the base station **104** selects the carrier **110** or **112**, is approximately 26 GHz. In alternative implementations, this central frequency is approximately 28 GHz or 39 GHz. The central frequency of the transmission band in low-frequency spectrum, within which the base station **104** selects the other carrier **110** or **112** according to these implementations, is approximately 700 MHz, 800 MHz, 900 MHz, or 1800 MHz. More generally, the transmission band in high-frequency spectrum can be separated from the transmission band in low-frequency spectrum by at least a certain bandwidth, e.g., 20 MHz, 200 MHz, 1 GHz. Further, in an example implementation, the high-frequency transmission band is disposed above 6 GHz, and the low-frequency transmission band is disposed below 6 GHz.
(16) With continued reference to FIG. **1**, the UE **102** can be equipped with processing hardware **120** that can include one or more general-purpose processors (e.g., CPUs) and a non-transitory

computer-readable memory storing instructions that the one or more general-purpose processors execute. Additionally or alternatively, the processing hardware **120** can include special-purpose processing units.

(17) The processing hardware **120** can include a channel status estimator **122**, a pathloss calculator **124**, and a NUL/SUL selector **126**. Each of the controllers **122**, **124**, and **126** can be implemented using any suitable combination of hardware, software, and firmware. In one example implementation, the controllers **122**, **124**, and **126** are sets of instructions that define respective components of the operating system of the UE **102**, and one or more CPUs execute these instructions to perform the corresponding functions. In another implementation, some or all of the controllers **122**, **124**, and **126** are implemented using firmware as a part of the wireless communication chipset.

(18) Similarly, the base station **104** can be equipped with processing hardware **130** that can include one or more general-purpose processors (e.g., CPUs) and a non-transitory computer-readable memory storing instructions that the one or more general-purpose processors execute. Additionally or alternatively, the processing hardware **130** can include special-purpose processing units. The processing hardware **130** can include a channel status estimator **132**, a pathloss calculator **134**, and a NUL/SUL selector **136**. Similar to the controllers **122**, **124**, and **126**, each of the controllers **132**, **134**, and **136** can be implemented using any suitable combination of hardware, software, and firmware.

(19) As discussed in more detail below, the processing hardware **130** of the base station **104** initially configures the UE **102** with the NUL **110** and SUL **112**. The UE **102** thus initially transmits data in the uplink direction over the NUL **110**. However, the channel status estimator **122** generates a metric indicative of the current condition of the NUL **110** and/or the SUL **112**, i.e., of how well the NUL **110** and/or the SUL **112** perform in terms of interference, traffic load, power level, etc. This metric can be any suitable quantitative indicator and is referred below as the “channel status metric.” The channel status estimator **122** in various implementations generates the metric periodically (e.g., after a certain amount of time, a certain number of transmitted frames) or in response to certain events, such as indications from upper layers that packets are lost or delayed. Moreover, the channel status estimator **122** in some implementations operates with the periodicity, or in accordance with conditions, specified by the base station **104**.

(20) The pathloss calculator **124** can determine pathloss of the NUL **110** and/or the SUL **112**. Similar to the channel status estimator **122**, the pathloss calculator **124** can calculate pathloss periodically, in response to events at the UE **102**, or in accordance with the commands from the base station **104**. The NUL/SUL selector **126** then can select the uplink carrier in accordance with the outputs of the channel status estimator **122** and the pathloss calculator **124**.

(21) As further discussed in more detail below, the base station **104** also monitors the performance of the NUL **110** and/or the SUL **112**, additionally to the UE **102** or as an alternative to the UE **102**. To this end, the components **132**, **134**, and **136** operate in a manner generally similar to that of the components **122**, **124**, and **126**. However, the components **132**, **134**, and **136** in some implementations determine whether the performance of the NUL **110** and the SUL **112** meets certain additional conditions or, conversely, the components **132**, **134**, and **136** in other implementations can enforce fewer conditions that the components **122**, **124**, and **126**.

(22) Next, FIG. **2** illustrates a messaging diagram **200** of an example scenario in which the NUL/SUL selector **126** (or another suitable component of the UE **102**) determines that that the UE **102** should switch from the NUL **110** to the SUL **112**, or vice versa in some cases. Of course, after the UE **102** transmits data over the SUL **112** for some time, the NUL/SUL selector **126** in some cases can determine that the UE **102** should switch back to the NUL **110**.

(23) At block **202**, the base station or gNB **104** can configure the UE **102** with two uplink carriers, e.g., the NUL **110** and the SUL **112**. As illustrated in FIG. **4**, the UE **102** at block **202** can perform a procedure associated with a protocol for controlling radio resources between the UE **102** and the

base station **104**, e.g., the reconfiguration procedure of the Radio Resource Control (RRC) sublayer of the protocol communication stack.
(24) In another implementation, the base station **104** configures the UE **102** with the first uplink carrier and the second uplink carrier via the RRCSetup message transmitted as a part of the RRC Connection Establishment procedure. In yet another implementation, the base station **104** configures the UE **102** with the first uplink carrier and the second uplink carrier via the RRCEstablishment message transmitted as a part of the RRC Re-Establishment procedure. In still another implementation, the base station **104** configures the UE **102** with the first uplink carrier and the second uplink carrier via the RRCResume message transmitted as a part of the RRC Connection Resume procedure.
(25) As illustrated in FIG. **4** (see message **422**), the UE **102** in implementations specifies, to the base station **104**, its capability to operate in particular unlicensed bands, licensed bands, or combinations of licensed and unlicensed bands. Thus, the processing hardware **130** of the base station **104** can select the first uplink carrier and the second uplink carrier in view of which the transmission bands are accessible to the hardware of the UE **102**. Further, when selecting the first uplink carrier and the second uplink carrier at block **202**, the base station **104** can consider channel status. The channel status estimator **132** can apply the same logic (discussed in more detail below) to generate a channel status metric as during the process of determining whether the UE **102** should switch between uplink carriers.
(26) With continued reference to FIG. **2**, the UE **102** initially transmits data in the uplink direction over the first uplink carrier, at block **210**. At block **230**, the NUL/SUL selector **126** can determine that the UE **102** should switch to the second uplink carrier. An example procedure for determining, at a user device, whether one uplink carrier is preferable to another uplink carrier is discussed below with reference to FIG. **5**.
(27) When the UE **102** determines that it should not switch to the other uplink carrier, the UE **102** can continue to transmit data in the uplink direction over the first uplink carrier, similar to block **210**. Otherwise, when the UE **102** determines that it should switch to the other uplink carrier, the UE **102** notifies the base station **104** at block **240**.
(28) In one example implementation, after the UE **102** switches to the second carrier, the UE **102** initiates a random access procedure on the newly selected carrier in order to inform the base station **104** of the switch. The base station **104** accordingly can allocate an uplink grant on the new uplink carrier, which the UE **102** can use in subsequent uplink transmissions.
(29) In another implementation or scenario, when the UE **102** determines that it should switch from the first uplink to the second uplink carrier at block **230**, the UE **102** already has an uplink grant to transmit uplink data over the first uplink carrier. In this case, at block **240** the UE **102** includes a MAC control element (including a logical channel ID, for example) in the uplink data to inform the UE **104** of the switch to the second uplink carrier. The base station **104** accordingly can allocate an uplink grant on the new uplink carrier, which the UE **102** can use in subsequent uplink transmissions. The UE **102** then can transmit data over the second uplink carrier, as illustrated in block **250**.
(30) FIG. **3** illustrates a messaging diagram **300** of an example scenario in which the NUL/SUL selector **136** (or another suitable component of the base station **104**) determines that that the UE **102** should switch from the NUL **110** to the SUL **112**, or vice versa in some cases. Blocks **302** and **310** in FIG. **3** are similar to blocks **202** and **210**, respectively, discussed above with reference to FIG. **2**.
(31) At block **320**, the UE **102** generates a first channel status metric for the first uplink carrier, a second channel status metric for the second uplink carrier, or both. Example techniques for generating these metrics are discussed with reference to FIG. **5** below. The UE **102** then provides these channel status metrics to the base station **104**. The UE **102** in some cases also can provide the calculated pathloss values to the base station **104**. In some implementations, the base station **104**

specifies the periodicity for these reports, at block **302**. Further, the base station **104** can specify, as a part of the configuration provided at block **302**, conditions for reporting the channel status metrics (and possibly other metrics) to the base station **104**. As one example, the base station **104** can specify that the UE **102** should report the channel status metrics only upon determining that the channel status metrics, pathloss calculations, etc. satisfy one or more conditions for switching to the second uplink carrier (see block **230** in FIG. **2**).
(32) In one implementation, the UE **102** sends the channel status metric(s) to the base station **104** as part of uplink control information. In another implementation, the UE **102** sends the channel status metric(s) to the base station **104** in a MAC control element (including a logical channel ID, for example). In yet another implementation, the UE **102** sends the channel status metric(s) to the base station **104** in a certain RRC message. For example, the UE **102** can transmit the channel status in the RRCMeasurementReport message during the measurement reporting procedure.
(33) In one implementation, the UE **102** determines whether the one or more channel status metrics, pathloss calculations, etc. satisfy the one or several conditions for switching to another uplink carrier, similar to block **230**. The UE **102** starts a timer at block **302** and, if the status metrics, pathloss calculations, etc. continue to satisfy the one or more conditions at block **320**, the UE **102** reports the channel status metric(s) to the base station. Further, the UE **102** also can start a timer after switching to another uplink carrier, and not switch back to the first carrier before the timer expires, as discussed in more detail below.
(34) At block **332**, the base station **104** can determine that the UE **102** should switch to the second uplink carrier. An example procedure for determining, at a base station, whether one uplink carrier is preferable to another uplink carrier is discussed below with reference to FIG. **5**.
(35) The base station **104** at block **342** can instruct the UE **102** to switch to the second uplink carrier. The UE **102** begins to transmit data over the second uplink carrier at block **240**.
(36) Referring to both FIGS. **1** and **2**, the UE **102** and the base station **104** can encounter a race condition when executing blocks **230** and **332**, respectively. In this case, the base station **104** can transmit a command to the UE **102** to change the PCell or switch the uplink carrier, whereas the UE **102** can determine that it should perform a different switching procedure. A timer in this case improves the probability that the UE **102** can comply with the command from the base station **104**.
(37) For clarity, FIG. **4** illustrates an example scenario **400** in which the base station **104** queries the user device **102** to determine which licensed and unlicensed bands the user device supports, and configures the two uplink carriers accordingly.
(38) According to the scenario **400**, the UE **102** and the base station **104** first synchronize communications over an uplink carrier. To this end, the UE **102** first transmits **402** a random access preamble to the base station **104**. The base station **104** transmits **404** a random access response **404**. Upon establishing access to the uplink carrier in this manner, the UE **102** transmits **410** an RRCSetupRequest message, receives **412** an RRCSetupComplete message in response, and transitions to the RRC_CONNECTED state.
(39) The base station **104** then queries the UE **102** by transmitting **420** a UECapabilityEnquiry message. The UE **102** responds with UECapabilityInformation that specifies one or more licensed bands, one or more unlicensed bands, and in some cases a combination of the one or more licensed bands and the one or more unlicensed bands in which the UE **102** can transmit over uplink carries.
(40) The base station **104** uses the capabilities reported via the UECapabilityEnquiry message to configure the NUL **110** and the SUL **112**, and transmits **430** the corresponding uplink carrier configuration to the UE **102** via an RRCReconfiguration message. The UE **112** configures the first uplink carrier and the second uplink carrier accordingly, and responds **432** with the RRCReconfigurationComplete message.
(41) Next, FIG. **5** illustrates an example method **500** for determining whether a user device should switch from one uplink carrier to another uplink carrier. The method **500** can be implemented in the modules **122**, **124**, and **126** (see FIG. **1**), for example, or another suitable component or a set of

components.
(42) For clarity, in the discussion of measurements and calculations at the UE **102** and the base station **104** below, the terms "first," "second," etc. can apply to channel status metrics and pathloss values as follows: the UE **102** can generate the first channel status metric for the first carrier (which in an example implementation is the NUL in a high-frequency transmission band) and the second channel status metric for the second carrier (which in this example implementation is the SUL in a low-frequency transmission band); and the base station **104** can generate the third channel status metric for the first carrier and the fourth channel status metric for the second carrier; the UE **102** can calculate the first pathloss for the first carrier and the second pathloss for the second carrier, based on the corresponding sounding signals from the base station **104**; and the UE base station **104** can calculate the third pathloss for the first carrier and the fourth pathloss for the second carrier, based on the corresponding sounding signals from the UE **102**.
(43) At block **502**, the UE **102** performs a measurement on one or both uplink carriers to generate respective channel status metrics. The UE **102** for example can generate a first channel status metric for the first uplink carrier and a second channel status metric for the second uplink carrier.
(44) In an example implementation, the UE **102** uses a moving average function, denoted as channel_status(i−1), to derive the channel status for the i-th measurement. Thus, channel_status (i)=a*channel_status(i−1)+(1−a)*measurement_result(i). The UE **102** can store a predefined value for the parameter a, or the UE **102** can receive the parameter a from the base station **104** at block **202** as a part of the configuration of the uplink carriers.
(45) In some implementations, the UE **102** measures the power level on each measurement occasion, which can be the number of OFDM symbols in a received frame. If the received power level is higher than a certain threshold, the UE **102** can set the measurement result to a certain first value (e.g. "1"). Otherwise, if the measured power level is lower than a certain threshold, the UE **102** can set the measurement result to a second value (e.g. "0"). The UE **102** can store a predefined value of the threshold, or the UE **102** can receive this threshold from the base station **104** at block **202** or **302** as a part of the configuration of the uplink carriers.
(46) In another implementation, the base station **104** allocates a time/frequency resource to the UE **102**, and specifies this time/frequency resource during configuration at block **202**. During the time period corresponding to the time/frequency resource, the base station **104** in this implementation does not perform any transmissions, so the UE **102** can measure the power level for the time/frequency tuple. This measurement does not take the transmission from the base station **104** into consideration, and thus yields a "clean" metric for the channel status in the proximity of the UE **102**. If the received power level is higher than a threshold, the UE **102** sets the measurement result to a first value (e.g. "1"). Otherwise, if the measured power level is lower than a threshold, the UE **102** sets the measurement result to a second value (e.g. "0"). Similar to the examples above, the UE **102** can store a predefined value of the threshold, or the UE **102** can receive this threshold from the base station **104** at block **202** or **302** as a part of the configuration of the uplink carriers.
(47) In yet another implementation, the base station **104** similarly allocates a time/frequency resource to the UE **102**, and specifies this time/frequency resource during configuration at block **202** or **302**. During the time period corresponding to the time/frequency resource, the base station in this implementation transmits a reference signal to user devices including the UE **102**. The UEs know the transmit power and time/frequency resources for the reference signal from system information, dedicated RRC messages, or pre-defined rules. The UE **102** in this case obtains the measurement result on the time/frequency resource by calculating the Signal-to-Interference ratio or the Reference Signal Received Quality (RSRQ). In some implementations, the measurement result can correspond to the inverted signal-to-interference ratio or inverted RSRQ.
(48) At block **504**, the UE **102** determines pathloss on one or both uplink carriers. In an example implementation, the UE **102** determines a first pathloss on the first uplink carrier and a second pathloss on second uplink carrier. The UE **102** can obtain the first and second pathloss from

reference signals which the base station **104** transmits. Examples of such reference signals include a cell reference signal, Channel Status Information Reference Signal (CSI-RS), and a demodulation reference signal.
(49) In various implementations, the UE **102** can determine that it should switch from one uplink carrier to another uplink carrier upon determining that one or more channel status metrics, pathloss calculations, etc. satisfy one or several conditions discussed next.
(50) For example, at block **506**, the UE **102** compares the channel status metric and/or the determined pathloss to corresponding thresholds. According to one implementation, the UE **102** determines that it should switch to the second uplink carrier if the first channel status metric is equal to exceeds a certain channel status threshold (a larger value of the channel status metric in this case corresponds to worse performance of the carrier), or if the first pathloss is equal to exceeds the corresponding pathloss threshold. For example, the first channel status metric and the first pathloss can correspond to the NUL **110**. Thus, when the NUL **110** is allocated within a high-frequency transmission band and the SUL is allocated within a low-frequency transmission band, and when the UE **102** experiences large pathloss, high interference, or high traffic load on the high-frequency carrier, the UE **102** can switch to the low-frequency carrier to improve coverage or channel connection.
(51) According to another example condition, the UE **102** switches from the second uplink carrier to the first uplink carrier when the first pathloss is less than or equal to a certain threshold and the first channel status metric is less than or equal to another threshold. Thus, when the NUL **110** is allocated within a high-frequency transmission band and the SUL is allocated within a low-frequency transmission band, and when the UE **102** detects low pathloss and low interference/traffic load on the high-frequency carrier, the UE **102** can switch to the high-frequency carrier to boost uplink transmission rate. As noted above, typically there is more transmission bandwidth in high-frequency spectrum, and thus using high-frequency spectrum to transmit data can increase data transmission rate.
(52) According to another example condition, the UE **102** switches from the second uplink carrier to the first uplink carrier if the first pathloss is less than a certain threshold, the first channel status metric is less than another threshold, and the second channel status metric exceeds or is equal to another threshold. This condition is similar to the preceding one, but the UE **102** in this case also considers the channel status metric for the second carrier. If the channel status metric of the second carrier is not sufficiently large, the UE **102** in this implementation does not switch to the first carrier, which can prevent frequent switching between two carriers.
(53) According to another example condition, the UE **102** switches from the second uplink carrier to the first uplink carrier if the first pathloss is less than or equal to a certain threshold and the first channel status metric is less than or equal to a sum of the second channel status metric and a certain offset. In other words, according to this condition, the difference between the first channel status metric and the second channel status metric must be sufficiently large.
(54) When the UE **102** determines at block that one or more conditions have been satisfied, the flow proceeds to block **508**. Otherwise, the flow returns to block **502**, and the UE **102** continues to transmit over the first uplink carrier. At block **508**, the UE **102** can start a timer to determine whether the one or more channel status metrics, pathloss calculations, etc. continues to satisfy the one or several conditions satisfied at block **506**, during a certain period of time. In another implementation, however, the flow proceeds directly from block **506** to block **514** when the one or more conditions have been satisfied.
(55) At block **510**, the UE **102** checks the same condition as in block **506**. If the channel status metrics, pathloss calculations, etc. satisfy the condition, the UE **102** checks whether the time has expired at block **512**. The method **500** terminates otherwise. The flow returns to block **510** if the timer has not yet expired. Otherwise, the flow proceeds to block **514**, where the UE **102** switches to another uplink carrier. Generally speaking, the mechanism of blocks **508-512** can prevent the “ping

pong” effect of continuously switching between two uplink carriers.
(56) In an alternative implementation, the UE **102** starts a timer after the UE **102** switching from the first uplink carrier to the second uplink carrier. The UE **102** does not switch back to the first carrier before the timer expires. Similar to the mechanism of blocks **508-512** discussed above, this technique prevents the ping pong effect of frequently switching between uplink carriers.
(57) In some cases, after the UE **102** switches from one uplink carrier to another uplink carrier, the UE **102** aborts the channel access procedure the UE **102** was performing on the previous carrier when the previous carrier is allocated in unlicensed transmission band, or the UE **102** aborts the random access procedure that the UE is performing on the previous carrier when the previous carrier is allocated in unlicensed transmission band.
(58) FIG. **6** illustrates an example method **600** for determining whether a user device should switch from one uplink carrier to another uplink carrier. The method **600** can be implemented in the modules **132**, **134**, and **136** (see FIG. **1**), for example, or another suitable component or a set of components.
(59) At block **602**, the base station **104** in various implementations can perform a measurement on an uplink carrier, perform measurements on both uplink carriers, and/or receive one or more channel status metrics from the UE **102**.
(60) When measuring power on the first uplink carrier, the base station **104** can determine that the measured power level is higher than a power threshold and set the measurement result is set to a first value (e.g. “1”). When the base station **104** determines that the measured power level is not higher than the power threshold, the UE **102** can set the measurement result is set to a second value (e.g. “0”). To generate the third channel status metric for the i-th measurement, the base station **104** can apply the formula channel_status (i)=a*channel_status(i−1)+(1−a)*measurement_result(i), similar to the UE **102** (see the discussion of block **502**).
(61) In an example scenario, the base station **104** derives a third pathloss and a third channel status on the first carrier, and calculates a fourth pathloss and a fourth channel status on the second carrier. To calculate the third pathloss, the UE **102** can transmit a sounding reference signal, and the base station **104** can determine the third pathloss based on the sounding reference single. When the base station **104** receives a channel status metric or a reference signal from UE **102**, the base station **104** determines whether the UE **102** should switch uplink carriers. Further, in some implementations, the base station **104** periodically checks whether the UE **102** should switch uplink carriers and, to this end, the base station **104** periodically performs measurements on the first uplink carrier and the second uplink carrier.
(62) At block **604**, the base station **104** compares one or more metrics, pathloss indications, etc. to respective thresholds. The base station **104** then can determine whether these comparisons satisfy one or more conditions for instructing the UE **102** to switch uplink carriers.
(63) One example condition is that the first channel status metric be greater than or equal to a certain channel status threshold. If the first channel status and the channel status threshold satisfy this condition, the base station **104** can instruct the UE to switch from the first uplink carrier to the second uplink carrier. This base station **104** may encounter this situation when high interference and/or traffic load occur on the first uplink carrier.
(64) According to another example condition, the third channel status metric is greater than a certain channel status threshold. The base station **104** in this case can instruct the UE **102** to switch from the first uplink carrier to the second uplink carrier. This situation may be due to high interference or high traffic load on the first uplink carrier. Unlike the previous scenario, the base station **104** here determines that the UE **102** should switch uplink carriers based on the assessment of a channel at the base station **104** rather than at the UE **102**.
(65) According to another example condition, the base station **104** determines that the first channel status metric is less than or equal to a certain threshold, and the third channel status metric is less than or equal to another threshold. In this scenario, the first uplink carrier (to which the first

channel status metric corresponds) is in a high-frequency transmission band, and therefore has higher bandwidth. The second uplink carrier accordingly is in a low-frequency transmission band, with lower bandwidth. The base station **104** in this case instructs the UE **102** to switch from the second uplink carrier to the first uplink carrier. This may occur when the interference/traffic load in the proximity of the UE **102** and the base station **104** becomes relatively low on the first uplink carrier. The base station **104** accordingly can instruct the UE **102** to switch the uplink carrier so as to increase the uplink data transmission rate.
(66) In another implementation, when the base station **104** determines that the first channel status metric is less or equal to than a certain channel status threshold, that the third channel status metric is less than or equal to another channel status threshold, and that the third pathloss is less than or equal to another threshold, the base station **104** instructs the UE **102** to switch from the second uplink carrier to the first uplink carrier.
(67) As illustrated in blocks **608-612**, when the metrics, pathloss indications, etc. satisfy one or more conditions discussed above, the base station **104** can start a timer. If the one or more conditions are still satisfied upon timer expiration (block **612**), the flow proceeds to block **614**. Otherwise, the method **600** terminates after block **610**. Similar to the discussion above with reference to FIG. **5**, using a timer at blocks **608-612** prevent the effect of ping-ponging between the two uplink carriers.
(68) At block **614**, the base station **104** transmits, to the UE **102**, a command to switch to the other uplink carrier. In one implementation, the base station **104** transmits a downlink control element (DCI) on Physical Downlink Control Channel (PDCCH) to instruct the UE **102** to perform carrier switching. According to another implementation, the base station **104** transmits to the UE **102** a MAC control element to instruct the UE **102** to perform carrier switching, similar to the examples above. In yet another implementation, the base station **104** transmits an RRC message to instruct the UE **102** to perform uplink carrier switching. For example, the base station **104** can transmit the channel status in an RRCReconfiguration message during the RRC reconfiguration procedure.
(69) Similar to the situations when the UE **102** determines that it should switch uplink carriers, the UE **102** can use a timer in the situations where the base station **104** instructs the UE **102** to switch uplink carriers. This timer can prevent the UE **102** from switching back to the other carrier before the timer expires and potentially creating a situation which the UE **102** frequently switches between the first uplink carrier and the second uplink carrier.
(70) For further clarity, FIG. **7** illustrates an example method **700** in a user device for configuring a base station with two uplink carriers and receiving data over these uplink carriers, and FIG. **8** illustrates an example method **800** in a user device for transmitting data over one of two uplink carriers. The methods **700** and **800** can be implemented in the UE **102** and the base station **104**, respectively, or in any other suitable devices.
(71) Referring first to FIG. **7**, the method **700** can begin at a step **702**, where the base station receives an indication of support bands in licensed and unlicensed spectrum from a user device. Alternatively, a base station can be pre-configured with certain transmission bands which each user device presumably can access. Thus, the method **700** in other implementations begins at block **704**.
(72) In any case, the base station at block **704** selects a first uplink carrier and a second uplink carrier for a user device. One or both uplink carriers can operate in unlicensed spectrum. At block **706**, the base station transmits the configuration of the two uplink carriers to the user device.
(73) The base station receives data via the first uplink carrier at block **708**. When the base station receives an indication that the user device switched to the other uplink carrier (block **710**), the base station receives further data over the other uplink carrier at block **712**.
(74) Further, the base station in some cases can determine one or more pathloss values and one or more channel status metrics at block **720**. The base station also can receive one or more pathloss values and one or more channel status metrics from the user device at block **722**. Using any suitable combination of these metrics, measurements, etc. generated at the base station and/or the

use device, the base station can determine whether the user device should switch to the other uplink carrier (block **724**). The base station can transmit a command to the user device, instructing the use device to switch uplink carriers (block **726**), after the base station determines at block **724** the user device should switch uplink carriers.

(75) Finally, the method **800** in a user device begins at block **802**, where the user device receives a configuration of a first uplink carrier and a second uplink carrier, with at least one of the first uplink carrier and the second uplink carrier allocated in unlicensed spectrum. The user device transmits over one of the uplink carriers at block **804**. After the user device detects a condition for switching uplink transmissions to the other carrier at block **806**, the user device transmits further data over the other carrier at block **808**.

(76) The following additional considerations apply to the foregoing discussion.

(77) A user device in which the techniques of this disclosure can be implemented (e.g., the UE **102**) can be any suitable device capable of wireless communications such as a smartphone, a tablet computer, a laptop computer, a mobile gaming console, a point-of-sale (POS) terminal, a health monitoring device, a drone, a camera, a media-streaming dongle or another personal media device, a wearable device such as a smartwatch, a wireless hotspot, a femtocell, or a broadband router. Further, the user device in some cases may be embedded in an electronic system such as the head unit of a vehicle or an advanced driver assistance system (ADAS). Still further, the user device can operate as an internet-of-things (IoT) device or a mobile-internet device (MID). Depending on the type, the user device can include one or more general-purpose processors, a computer-readable memory, a user interface, one or more network interfaces, one or more sensors, etc.

(78) Certain embodiments are described in this disclosure as including logic or a number of components or modules. Modules may can be software modules (e.g., code stored on non-transitory machine-readable medium) or hardware modules. A hardware module is a tangible unit capable of performing certain operations and may be configured or arranged in a certain manner. A hardware module can comprise dedicated circuitry or logic that is permanently configured (e.g., as a special-purpose processor, such as a field programmable gate array (FPGA) or an application-specific integrated circuit (ASIC)) to perform certain operations. A hardware module may also comprise programmable logic or circuitry (e.g., as encompassed within a general-purpose processor or other programmable processor) that is temporarily configured by software to perform certain operations. The decision to implement a hardware module in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software) may be driven by cost and time considerations.

(79) When implemented in software, the techniques can be provided as part of the operating system, a library used by multiple applications, a particular software application, etc. The software can be executed by one or more general-purpose processors or one or more special-purpose processors.

(80) Upon reading this disclosure, those of skill in the art will appreciate still additional alternative structural and functional designs for configuring and using multiple uplink carriers in a cell through the disclosed principles herein. Thus, while particular embodiments and applications have been illustrated and described, it is to be understood that the disclosed embodiments are not limited to the precise construction and components disclosed herein. Various modifications, changes and variations, which will be apparent to those of ordinary skill in the art, may be made in the arrangement, operation and details of the method and apparatus disclosed herein without departing from the spirit and scope defined in the appended claims.

(81) The following list of aspects reflects a variety of the embodiments explicitly contemplated by the present disclosure.

(82) Aspect 1. A method in a base station for configuring uplink communications in an unlicensed spectrum comprises selecting, by processing hardware, a first uplink carrier and a second uplink carrier for a cell, respectively, with at least one of the first uplink carrier and the second uplink

carrier operating within one or more unlicensed bands a user device supports; transmitting, by the processing hardware to the user device, configuration data for the first uplink carrier and the second uplink carrier, including transmitting an indication that one of the first uplink carrier and the second uplink carrier is a normal uplink carrier (NUL), and the other one of the first uplink carrier and the second uplink carrier is a supplementary uplink carrier (SUL), where the user device is configured to initially transmit via the NUL; and receiving uplink data via the second uplink carrier upon the user device switching uplink transmissions from the first uplink carrier to the second uplink carrier.

(83) Aspect 2. The method of aspect 1, further comprising, prior to receiving uplink data via the second uplink carrier: receiving, by the processing hardware from the user device, a notification that the user device switched from the first uplink carrier to the second uplink carrier.

(84) Aspect 3. The method of aspect 2, where receiving the notification includes: allocating, by the processing hardware, an uplink grant to the user device in response to the user device initiating a random access procedure on the second uplink carrier, and transmitting, by the processing hardware, the uplink grant to the user device.

(85) Aspect 4. The method of aspect 2, where receiving the notification includes receiving the notification in a control element at a media access control (MAC) layer, transmitted by the user device over the first uplink carrier.

(86) Aspect 5. The method of aspect 2, further comprising: calculating, by the processing hardware, a channel status indicative of a current condition of the second uplink carrier; selecting, by the processing hardware and based on the calculated channel status metric, the first uplink carrier for the user device; and transmitting a command to the user device to switch from the second uplink carrier back to the first uplink carrier

(87) Aspect 6. The method of aspect 1, further comprising: determining, by the processing hardware, that the user device should switch from the first uplink carrier to the second uplink carrier; and transmitting, by the processing hardware to the user device, a command to the user device to switch from the first uplink carrier to the second uplink carrier.

(88) Aspect 7. The method of aspect 6, where determining that the user device should switch from the first uplink carrier to the second uplink carrier includes at least one of: (i) receiving, from the user device, a first channel status metric indicative of a current condition of the first uplink carrier, (ii) receiving, from the user device, a second channel status metric indicative of a current condition of the second uplink carrier, (iii) receiving, from the user device, a first sounding reference signal indicative of a first pathloss, at the first uplink carrier, (iv) receiving, from the user device, a second sounding reference signal indicative a third pathloss, at the second uplink carrier, (v) calculating, at the base station, a third channel status metric indicative a current condition of the first uplink carrier, or (vi) calculating, at the base station, a fourth channel status metric indicative a current condition of the second uplink carrier.

(89) Aspect 8. The method of aspect 7, where calculating the third channel status or the fourth channel status includes measuring, for a corresponding carrier, one of: (i) an amount of interference on the carrier, (ii) an amount of traffic on the carrier, (iii) a power level, or (ii) a moving average of power levels.

(90) Aspect 9. The method of aspect 7, further comprising transmitting, by the processing hardware to the user device, an indication of a frequency and/or conditions for transmitting the first channel status and/or the second channel status metric.

(91) Aspect 10. The method of any of aspects 7-9, where determining that the user device should switch from the first uplink carrier to the second uplink carrier includes comparing the first channel status metric to a first channel status threshold.

(92) Aspect 11. The method of any of the aspects 7-9, where determining that the user device should switch from the first uplink carrier to the second uplink carrier includes comparing the third channel status metric to a third channel status threshold value.

(93) Aspect 12. The method of any of the aspects 7-9, where determining that the user device

should switch from the first uplink carrier to the second uplink carrier includes comparing the first channel status and the third channel status to respective channel status thresholds.
(94) Aspect 13. The method of any of the aspects 7-9, where determining that the user device should switch from the first uplink carrier to the second uplink carrier includes: comparing the first channel status metric to the first channel status threshold, comparing the third channel status metric to the third channel status threshold, and comparing the first sounding reference signal to a pathloss threshold.
(95) Aspect 14. The method of any of the aspects 7-9, where determining that the user device should switch from the first uplink carrier to the second uplink carrier includes: comparing the second channel status metric to the second channel status threshold, comparing the fourth channel status metric to the fourth channel status threshold, and comparing the second sounding reference signal to a pathloss threshold.
(96) Aspect 15. The method of aspect 6, where transmitting the command to the user device includes transmitting the command in one of (i) a downlink control element (DCI), (ii) a media access channel (MAC) control element, or (iii) a message of a protocol for controlling radio resources between the user device and the base station.
(97) Aspect 16. The method of aspect 6, further comprising: transmitting, by the processing hardware, a command to the user device to prevent the user device from switching back to the first uplink carrier during a certain period of time.
(98) Aspect 17. The method of any of the preceding aspects, where: selecting the first uplink carrier includes selecting the first uplink carrier in a high-frequency transmission band, and selecting the second uplink carrier includes selecting the second uplink carrier in a low-frequency transmission band, where respective central frequencies of the high-frequency transmission band and the low-frequency transmission band are separated by at least 20 MHz.
(99) Aspect 18. The method of any of the preceding aspects, wherein selecting the first uplink carrier and the second uplink carrier includes selecting both the first uplink carrier and the second uplink carrier from within the one or more unlicensed bands.
(100) Aspect 19. The method of any of the preceding aspects, further comprising prior to selecting: receiving, from the user device, an indication of the one or more unlicensed bands the user device supports.
(101) Aspect 20. The method of any of the preceding aspects, further comprising prior to selecting: receiving, from the user device, an indication of one or more licensed bands the user device supports.
(102) Aspect 21. The method of any of the preceding aspects, where transmitting the configuration data to the user device includes: transmitting the configuration data during one of (i) a procedure for establishing a connection associated with a protocol for controlling radio resources between the base station and the user device, (ii) a procedure for re-establishing a connection associated with the protocol, and (iii) a procedure for resuming a connection associated with the protocol.
(103) Aspect 22. A method in a user device for uplink communications comprises: receiving, by processing hardware from a base station for a certain cell, a configuration of a first uplink carrier and a second uplink carrier, where at least one of the first uplink carrier and the second uplink carrier are allocated in an unlicensed spectrum; receiving, by the processing hardware, an indication that the first uplink carrier is a normal uplink carrier (NUL), and the second one uplink carrier is a supplementary uplink carrier (SUL); transmitting, by the processing hardware, data over the NUL; detecting a condition for switching uplink transmissions from the NUL to the SUL; and in response to detecting the condition, transmitting further data over the SUL.
(104) Aspect 23. The method of aspect 22, where detecting the condition comprises: determining, by the processing hardware, a first pathloss on the NUL; and determining, by the processing hardware, a first channel status metric indicative of a current condition of the NUL.
(105) Aspect 24. The method aspect 23, where detecting the condition includes comparing the first

pathloss to a first pathloss threshold and the first channel status metric to a channel status threshold.
(106) Aspect 25. The method of aspect 23, further comprising: determining, by the processing hardware, a second pathloss on the NUL; where detecting the condition includes comparing the first pathloss to a first pathloss threshold, the first channel status metric to a channel status threshold, and the second pathloss to a second pathloss threshold.
(107) Aspect 26. The method of aspect 23, where calculating the first channel status metric includes measuring, for the first uplink carrier, one of: (i) an amount of interference on the carrier, (ii) an amount of traffic on the carrier, (iii) a power level, or (ii) a moving average of power levels.
(108) Aspect 27. The method of aspect 22, where detecting the condition includes receiving a command from the base station to switch from the NUL to the SUL.
(109) Aspect 28. The method of aspect 22, further comprising in response to detecting the condition: transmitting, by the processing hardware to the base station, a notification that the user device switched from the NUL to the SUL.
(110) Aspect 29. The method of aspect 28, where transmitting the notification includes: initiating, by the processing hardware, a random access procedure on the second uplink carrier, and receiving, from the base station, an uplink grant in response to the initiated random access procedure.
(111) Aspect 30. The method of aspect 28, where transmitting the notification includes transmitting the notification in a control element at a MAC layer, over the NUL.
(112) Aspect 31. The method of aspect 28, further comprising in response to transmitting the notification: receiving, by the processing hardware, a command from the base station to switch from the SUL back to the NUL.
(113) Aspect 32. The method according to any of the aspects 22-27, further comprising: transmitting, by the processing hardware, a notification that the user device switched from the NUL to the SUL.
(114) Aspect 33. The method according to any of the aspects 23-28, further comprising: preventing the user device from switching from the second uplink carrier to the first uplink carrier for a certain period of time after the transmitting further data.

Claims

1. A method in a base station for configuring uplink communications in an unlicensed spectrum, the method comprising: selecting a first uplink carrier and a second uplink carrier for a cell, respectively, with at least one of the first uplink carrier and the second uplink carrier operating within one or more unlicensed bands a user device supports; transmitting, to the user device, configuration data for the first uplink carrier and the second uplink carrier, including transmitting (i) an indication that one of the first uplink carrier and the second uplink carrier is a normal uplink carrier (NUL), and the other one of the first uplink carrier and the second uplink carrier is a supplementary uplink carrier (SUL), and (ii) a threshold value of a power level the UE measures on the first uplink carrier and/or the second uplink carrier, wherein the user device is configured to initially transmit via the NUL; receiving, from the user device, a notification that the user device switched, in response to a determination at the user device, from the first uplink carrier to the second uplink carrier; and receiving uplink data via the second uplink carrier upon the user device switching uplink transmissions from the first uplink carrier to the second uplink carrier.
2. The method of claim 1, wherein receiving the notification includes: allocating an uplink grant to the user device in response to the user device initiating a random access procedure on the second uplink carrier, and transmitting the uplink grant to the user device.
3. The method of claim 1, further comprising: calculating a channel status indicative of a current condition of the second uplink carrier; selecting, based on the calculated channel status metric, the first uplink carrier for the user device; and transmitting a command to the user device to switch from the second uplink carrier back to the first uplink carrier.

4. The method of claim 1, further comprising: transmitting, to the user device, an indication of a frequency and/or conditions for generating, at the user device, and transmitting to the base station, a first channel status metric indicative of a current condition of the first uplink carrier and/or a second channel status metric indicative of a current condition of the second uplink carrier.
5. The method of claim 1, further comprising: transmitting a command to the user device to prevent the user device from switching back to the first uplink carrier during a certain period of time.
6. The method of claim 1, wherein: selecting the first uplink carrier includes selecting the first uplink carrier in a high-frequency transmission band, and selecting the second uplink carrier includes selecting the second uplink carrier in a low-frequency transmission band, wherein respective central frequencies of the high-frequency transmission band and the low-frequency transmission band are separated by at least 20 MHz.
7. The method of claim 1, further comprising prior to selecting: receiving, from the user device, at least one of: an indication of the one or more unlicensed bands the user device supports, or an indication of one or more licensed bands the user device supports.
8. A method in a user device for uplink communications, the method comprising: receiving, from a base station for a certain cell, a configuration of a first uplink carrier and a second uplink carrier, wherein at least one of the first uplink carrier and the second uplink carrier are allocated in an unlicensed spectrum; receiving an indication that the first uplink carrier is a normal uplink carrier (NUL), and the second one uplink carrier is a supplementary uplink carrier (SUL); transmitting data over the NUL; determining that a pathloss on the SUL satisfies a condition for switching uplink transmissions from the NUL to the SUL; in response to the detecting of the condition, transmitting further data over the SUL; and transmitting, to the base station and in response to the detecting of the condition, a notification that the user device switched from the NUL to the SUL.
9. The method of claim 8, further comprising: determining a first pathloss on the NUL; and determining a first channel status metric indicative of a current condition of the NUL.
10. The method claim 9, wherein detecting the condition includes comparing the first pathloss to a first pathloss threshold and the first channel status metric to a channel status threshold.
11. The method of claim 9, wherein: the pathloss on the SUL is a second pathloss; the method further comprising: comparing the first pathloss to a first pathloss threshold, the first channel status metric to a channel status threshold, and the second pathloss to a second pathloss threshold.
12. The method of claim 8, wherein transmitting the notification includes: initiating a random access procedure on the second uplink carrier, and receiving, from the base station, an uplink grant in response to the initiated random access procedure.
13. The method of claim 12, further comprising in response to transmitting the notification: receiving a command from the base station to switch from the second uplink carrier back to the first uplink carrier.
14. The method according to claim 8, further comprising: preventing the user device from switching from the SUL to the NUL for a certain period of time after the transmitting further data.
15. A user device (UE) comprising: a transceiver; and processing hardware; the UE configured to: receive, from a base station for a certain cell, a configuration of a first uplink carrier and a second uplink carrier, wherein at least one of the first uplink carrier and the second uplink carrier are allocated in an unlicensed spectrum; receive, an indication that the first uplink carrier is a normal uplink carrier (NUL), and the second one uplink carrier is a supplementary uplink carrier (SUL); transmit data over the NUL; determine that a pathloss on the SUL satisfies a condition for switching uplink transmissions from the NUL to the SUL; in response to the detecting of the condition, transmit further data over the SUL; and transmit, to the base station and in response to the detecting of the condition, a notification that the user device switched from the NUL to the SUL.
16. The UE of claim 15, further configured to: determine first pathloss on the NUL, wherein the pathloss on the SUL is a second pathloss; determine a first channel status metric indicative of a

current condition of the NUL; and to detect the condition for switching uplink transmissions from the NUL to the SUL, the UE is configured to compare the first pathloss to a first pathloss threshold and the first channel status metric to a channel status threshold.